RVS COLLEGE OF ARTS AND SCIENCE (AUTONOMOUS), SULUR

SCHOOL OF COMPUTER SCIENCE - UG

II B.Sc. B.Sc. Artificial Intelligence and Machine Learning (2025 Batch)

Subject Code: **Subject: Data Analysis and Visualization using Excel**

PROGRAM LIST

1. **Demonstrate the usage of Excel String functions LENGTH, MID, RIGHT, LEFT using the below excel sheet.**

	A	B	C	D	E
1	Part #	Length	Extract first two characters	Extract last two characters	Extract middle characters
2	9C0K904	7			
3	ABCPC1212K	10			
4	INV201415011	12			
5	3G7R230	7			

2. **Illustrate the way to perform SUMPRODUCT function in the below excel sheet.**

	A	B	C
1	Product	Quantity	Price
2	Computer	2	40000
3	Keyboard	4	1200
4	Mouse	4	750
5	Printer	2	5000

3. **Illustrate the usage of the IF condition in the below excel sheet.**

	A	B	C	D	E	F	G
1	Employee Name	Hire Date	Years	Status	Job Rating	Bonus Eligibility - One Condition	Bonus Eligibility - Two Condition
2	Arjun, Puri	04-Jun-93	30	Half-Time	3		
3	Kunal, Kashyap	18-Feb-94	29	Full Time	2		
4	Bose, Anirban	04-May-89	34	Contract	4		
5	Mendoza, Bobby	04-Jan-97	26	Hourly	5		

4. **Apply the Conditional formatting techniques for the below excel sheet, Highlight the Dairy Product, Highlight the Product, sold on last month, sales amount should be greater than Rs.500.**

	A	B	C	D
1	Date	Product	Region	Amount
2	9-Apr-15	Dairy	Gujarat	₹ 1,148.00
3	26-May-15	Produce	Gujarat	₹ 1,530.00
4	7-Dec-15	Produce	Gujarat	₹ 1,423.50
5	5-Nov-15	Dairy	Maharashtra	₹ 192.10
6	12-Jul-15	Dairy	Maharashtra	₹ 351.00
7	2-Jun-15	Grain	Maharashtra	₹ 560.40

5. **Calculate the Depreciation value for the below worksheets**

	A	B
1	Asset Price	10,00,000
2	Salvage Value	5,90,490
3	Life in Years	5

6. **Create a line chart for the following data**

	A	B
1	Month	Sales
2	Jan-2013	15,92,398
3	Feb-2013	15,97,197
4	Mar-2013	16,66,080
5	Apr-2013	24,84,340
6	May-2013	26,69,994
7	Jun-2013	50,81,937

7. **Create a column chart for the following data**

	A	B	C	D	E	F	G	H	I
1	Home Products								
2	World-wide Sales - Million								
3									
4		Jan	Feb	Mar	Apr	May	Jun	Total	% of Total
5	Asia	80	130	125	130	140	180	785	33.1%
6	Europe	60	80	80	100	90	100	510	21.5%
7	Africa	110	120	110	120	120	130	710	29.9%
8	Latin America	40	60	70	60	60	80	370	15.6%

8. **Create Bar Chart for the following data**

	A	B
1	Home Products	
2	World-wide Profits - Million	
3		Profits
4	India	80
5	USA	60
6	China	110
7	Sri Lanka	10
8	Germany	120
9	Turkey	30
10	Iran	44
11	Australia	78
12	Russia	54
13	Brazil	38
14	South Africa	(25)
15	New Zealand	67
16	Bangladesh	40
17	Total	706
18	Average	54

9. **Given below is the height and weight of a group of 8 people. Plot this on one single chart**

Person	Weight	Height
A	77	1.72
B	69	1.72
C	100	1.78
D	75	1.65
E	93	1.76
F	86	1.55
G	89	1.64
H	75	1.77

10. **Given here is a relationship between car density per thousand population and pollution levels in various cities, in ascending order of car density. Plot a chart to show the relationship**

Car Density per 1000 population	Pollution Levels
15.0	20%
15.6	20%
15.9	21%
16.4	21%
18.0	22%
19.1	22%
19.3	23%

11. Given below are some investment details. Calculate the returns on them

Date	Amount			
1-Jan-11	100,000			
8-Aug-11	20,000			
5-Jan-12	400,000		IRR	11.66%
12-Mar-13	120,000		XIRR	12.51%
20-Jun-14	80,000			
30-Jan-15	75,000			
1-Jan-16	-1,201,000			

12. If a person starts saving Rs 100,000 today, and continue to do it for the next 10 years, when she retires, what is her investment corpus going to be, assuming a 12% return? (use future value formula)

	A	B	C	D
4				
5	**Year**	**Investment**	**Total**	**No.of Years Invested**
6	1			
7	2			
8	3			
9	4			
10	5			
11	6			
12	7			
13	8			
14	9			
15	10			
16	Likely yearly returns			
17	**FV of the savings**		-	

13. Assume a person pays taxes on his income at 30%, and there is a surcharge applicable based on certain conditions. How can we consider the various scenarios.

	A	B	C	D	E
1	**Tax Burden**			**Tax Due**	
2	**Year**	**2015**			**3,00,000**
3	**Surcharge**	**0%**		**1%**	**3,10,000**
4	**Taxable Income**	**10,00,000**		**2%**	**3,20,000**
5	**Tax Rate**	**30%**		**3%**	**3,30,000**

14. Ajay has taken a loan of Rs 17 lakhs to buy a car, and is expected to pay a 14.5% interest on it over 5 years. Calculate the equated monthly installments (EMIs).

	A	B
1	**Car Loan**	**17,00,000**
2	**Interest Rate**	**14.75%**
3	**Tenure (Years)**	**5**

15. Demonstrate the COUNT () and Countif() related functions using the below excel sheets .

	A	B	C	D	E
1	FinShiksha				
2					
3					
4			Count	CountA	CountBlank
5	123				
6	RVS				
7	FinShiksha				
8	213421				
9					
10					

	A	B	C	D	E	F	G	H
1	FinShiksha							
2								
3	COUNTIF:							
4	Employee Name	Hire Date	Years	Status	Job Rating			
5	Roshan	04-Jun-93	30	Half-Time	3		Total Full Time	
6	Shilpa	18-Feb-94	29	Full Time	2		Total Half-Time	
7	Gautam	04-May-89	34	Contract	4		Total Contract	
8	Ajay	04-Jan-97	26	Hourly	5		Years >=15	
9	Vinay	23-Jun-90	33	Full Time	4		Years <15	
10	Neeti	28-Sep-93	30	Half-Time	2		Ratings = 5	
11	Deshpande	14-Oct-89	34	Full Time	5		Ratings >=3	
12	Arjun	30-Jul-95	28	Full Time	3			
13	Misbah	30-May-89	34	Full Time	1			
14	Sohail	25-Jan-02	21	Full Time	3			
15	Inder	25-Sep-98	25	Full Time	2			

16. Illustrate the way to perform SUMIF() & AVERAGE IF() functions in the below excel sheets.

	A	B	C	D	E	F	G
1	FinShiksha						
2	SUMIF:						
3	Category	Food	Sales				
4	Vegetables	Tomatoes	2300				
5	Vegetables	Celery	5500				
6	Fruits	Oranges	800				
7		Butter	400				
8	Vegetables	Carrots	4200				
9	Fruits	Apples	1200				
10		Curd	400				
11	Formula	Description					
12		Sum of the sales of all foods in the "Fruits" category.					
13		Sum of the sales of all foods in the "Vegetables" category.					
14		Sum of the sales of all foods that do not have a category specified.					
15							

	A	B	C	D	E	F
1	FinShiksha					
2						
4	Average If:					
5						
6	Name	Department	State	Hours	Rate	
7	Roshan	Finance	Raj	59.6	40.0	
8	Shilpa	Sales	TN	68.6	23.7	
9	Gautam	Executive	MP	40.0	17.9	
10	Ajay	Executive	Raj	29.3	13.1	
11	Vinay	Sales	TN	62.9	47.9	
12	Neeti	IT	Mah	87.6	23.0	
13	Deshpande	Finance	Goa	33.2	24.1	
14	Arjun	HR	Kar	87.3	19.7	
15	Misbah	Finance	Ker	96.3	35.9	
16	Sohail	Finance	Raj	85.3	24.1	
17	Inder	HR	Delhi	11.9	32.1	
18						
19	Average rate for RAJ					
20	Average rate for MP					
21						

17. Demonstrate the Exact VLOOKUP using the below Excel sheet.

FinShiksha

RateTable	
Excellent	99
Very Good	92
Good	85
Satisfactory	78
Fair	71
Poor	65
Fail	50

Name	Rating	Numerical Score
Babbitt, George	Fair	71
Checker, Charles	Satisfactory	78
Belli, Melvin	Satisfactory	78
Bench, John	Good	85
Bickle, Travis	Poor	65
Martinet, Jean	Satisfactory	78
Cabot, Sebastian	Very Good	92
Carson, Kit	Fail	50

18. Assume a company with following sales data from states across India. Represent it in a format where if someone selects a state, they get a chart displayed for that only.

Name	2,010	2,011	2,012	2,013
Maharashtra	22,246	33,082	42,747	58,429
Rajasthan	26,114	36,486	47,620	50,891
Tamil Nadu	45,419	54,686	56,527	59,243
Karnataka	26,431	33,576	40,719	37,823
Kerala	20,826	34,077	44,197	43,012
Andhra Pradesh	44,361	53,058	55,891	60,029
Madhya Pradesh	28,868	35,743	34,201	31,003
Sikkim	40,377	52,198	56,376	59,666
Gujarat	43,511	48,467	56,150	57,666
Assam	26,905	33,565	41,276	59,938

19. Write down the steps for create Item wise sales Product wise revenues in each city, for each distributor and for each month in Column Chart using Pivot table concept.

FinShiksha

Region	City	Distributor	Product	Sales	Month
Tamil Nadu	Coimbatore	Alpha	Ketchup	1,54,700	Jan-15
Tamil Nadu	Chennai	Beta	Glucose Buscuits	3,32,100	Jan-15
Tamil Nadu	Coimbatore	Alpha	Ketchup	1,03,600	Jan-15
Tamil Nadu	Chennai	Gamma	Jam	2,96,000	Jan-15
Tamil Nadu	Chennai	Beta	Ketchup	3,36,100	Jan-15
Tamil Nadu	Coimbatore	Alpha	Chocolate Buscuits	3,93,700	Jan-15
Tamil Nadu	Chennai	Alpha	Ketchup	1,98,100	Feb-15
Karnataka	Bangalore	Gamma	Glucose Buscuits	3,69,700	Mar-15
Karnataka	Mysore	Beta	Jam	1,66,700	Mar-15
Karnataka	Bangalore	Alpha	Glucose Buscuits	2,84,000	Apr-15
Tamil Nadu	Coimbatore	Alpha	Glucose Buscuits	3,70,200	Jan-15
Tamil Nadu	Chennai	Beta	Glucose Buscuits	3,59,100	Jan-15
Karnataka	Mysore	Gamma	Jam	3,21,900	Feb-15
Tamil Nadu	Coimbatore	Beta	Bread	2,32,900	Apr-15

20. Calculate the statistical parameters in returns for the given data.

	A	B
1	Date	Nifty
2	01-Mar-15	8492.3
3	31-Mar-15	8491
4	01-Apr-15	8586.25
5	06-Apr-15	8659.9
6	10-Apr-15	8780.35
7	13-Apr-15	8834
8	17-Apr-15	8606
9	23-Apr-15	8398.3
10	27-Apr-15	8213.8
11	30-Apr-15	8181.5
12	04-May-15	8331.95
13	08-May-15	8191.5
14	14-May-15	8224.2
15	21-May-15	8421
16	22-May-15	8458.95
17	25-May-15	8370.25
18	28-May-15	8319
19	29-May-15	8433.65
20	01-Jun-15	8433.4
21	05-Jun-15	8114.7
22	08-Jun-15	8044.15
23	12-Jun-15	7982.9
24	15-Jun-15	8013.9
25	17-Jun-15	8091.55
26	23-Jun-15	8381.55
27	29-Jun-15	8318.4